Phase Transitions and their Analogy to Bifurcations

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Abstract

The almost startling similarities between phase transitions and bifurcations are studied. The critical behaviour of both systems are studied independantly and then compared. We find that a first order phase transition(second order phase transition) is analogous to a modified saddle-node (subcritical pitchfork) bifurcation and attempt to generalise this treatment.

Introduction

Phase transitions are quite striking when first encountered - the generally smooth laws of physics seem to somehow, suddenly give in to discontinuities. There are precious few other topics in which such abrupt changes are seen - bifurcations in dynamical systems being one of them. This is an attempt to understand them - first separately, by their own theories and then by comparing the resuls.

1 Critical Point Behaviour of Phase Transitions

First we will look at how the Ising model in particular, behaves around its critical points and then, using Landau theory, paint a general picture for all phase transitions.

1.1 The Ising Model

First developed to explain ferromagnetism in statistical mechanics, the Ising model today is a barometer for a staggering number of applications:

The Ising model consits of N sites in a d-dimensional lattice; each site occupied by a quantum spin (that can be in either spin up state, $s_i = +1$ or spin down state $s_i = -1$). An external magnetic field B confers an energetic advantage to spins in $s_i = +1$; there is also an interaction energy between the spins (corresponding to quantity J). The energy E[1] and the partition function Z are given thus:

$$E = -B\sum_i s_i - J\sum_{<ij>} s_i s_j \tag{1}$$

$$Z = \sum_{s_i} e^{E[s_i]} \tag{2}$$

The average spin or magnetisation, m, is given by:

$$m = \frac{1}{N}\sum_i < s_i >= \frac{1}{N\beta}\frac{\partial log Z}{\partial B} \tag{3}$$

[1]The second summation term is over all the nearest neigbours for each spin i

Upon simplification by mean field theory (that replaces the interaction term with a contribution towards the effective magnetic field), we have:

$$B_{eff} = B + Jq^2 m \tag{4}$$

$$E_{mf} = \frac{1}{2} JNqm^2 - B_{eff} \sum_i s_i \tag{5}$$

$$Z = e^{\frac{1}{2}\beta JNqm^2} 2^N cosh^N \beta B_{eff} \tag{6}$$

Solving for m, we get

$$m = tanh(\beta B + \beta Jqm) \tag{7}$$

The behaviour is separately analysed for vanishing and non-zero exernal fields.

1.1.1 Critical Behaviour at 0 Field:

There exists a critical temperature, T_c given by

$$k_B T_c = Jq \tag{8}$$

1. For $T \leq T_c$: There are 3 solutions to (eq.7): $\pm m = m_0$ which are stable states and $m = 0$, an unstable state.

2. For $T \geq T_c$: Only $m = 0$ is the stable state hereon.

This is a second order phase transition.[3]

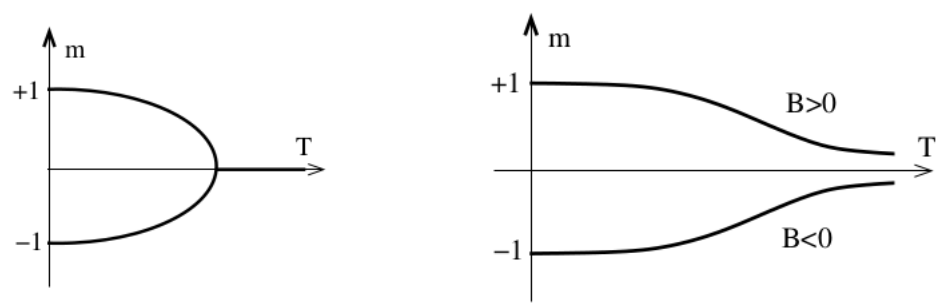

Figure 1: Magnetisation as a function of temperature with (a)LHS: $B = 0$ (b)RHS: $B \neq 0$

1.1.2 Critical Behaviour at $B \neq 0$:

There is no discontinuity in m with T, as is evident from the phase diagram. However, it is not apparent that as $B \to 0$ as the system can be viewed as a perturbation to the $B = 0$ case, there are actually 3 solutions to (eq. 7) to both $B \to 0^-$ and $B \to 0^+$ branches.
Only one of them is stable and this is the state that persists as $|B|$ increases. The other two are metastable and unstable respectively. (This is understood better in the Landau approach). This is a first order phase transition.

[3]In an n-th order phase transition, there is a discontinuity in the nth partial derivative of a potential (like G, Gibbs free energy or F, Helmholtz free energy).

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Around $T \to T_c$, Taylor expanding (eq. 7) and substituting $Jq = k_B T_c$

$$tanh(x) \approx x - \frac{1}{3}x^3 + \dots \tag{9}$$

$$\implies m \approx (\beta B + \frac{T_c}{T}m) - \frac{1}{3}\left(\beta B + \frac{T_c}{T}m\right)^3 \tag{10}$$

By analysing this, we can obtain the following critical exponents:

$$B = 0, T \to T_c^- \quad : \quad m \sim (T_c - T)^{\frac{1}{2}} \tag{11}$$

$$T = T_c, B \to 0 \quad : \quad m \sim B^{\frac{1}{3}} \tag{12}$$

$$B = 0, T \to T_c \quad : \quad \chi \sim (T_c - T)^{(} - 1) \tag{13}$$

$\chi \left(= N \frac{\partial m}{\partial B}\big|_T\right)$ here is the magnetic susceptibility.

Taking the additional limit $B \to 0$: we only consider the first order term in B; neglecting it altogether in the cubic term.

$$m = (\beta B + \frac{T_c}{T}m) - \frac{1}{3}\left(\frac{T_c}{T}m\right)^3 \tag{14}$$

$$\beta B - \left(\frac{T - T_c}{T}\right)m - \frac{1}{3}\left(\frac{T_c}{T}\right)m^3 = 0 \tag{15}$$

Therefore, upon defining $\theta = \left(\frac{T-T_c}{T}\right)$ and $h = \frac{B}{2Jq}$, we get the equation of state to be:

$$h - \theta m - \frac{1}{3}m^3 = 0 \tag{16}$$

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We have thus seen the two different kinds of phase transitions that occur in the Ising Model and how they behave around the critical points in each case. It was seen that the Van der Waal's equation of state for gasses too behaves similiarly , motivating a generalised treatment of phase transitions.

1.2 Landau Theory

The Landau approach to phase transitions developed because of the need to look at them in a unified langauge. It is based on free energy.

We identify an order paramter m, that is non-zero for $T < T_c$ but vanishes otherwise and obtain the Helmholtz free energy as a function of it $F(T; m)$. At a fixed temperature, minimising the free energy, i.e.

$$\frac{\partial F}{\partial m} = 0$$

$\implies$ we are in equilibrium; we can thus obtain m as a function of T.[4] We apply this to simplified first and second order phase transitions.

[4]The tradeoff for its relative simplicity is a reduced accuracy; often the Landau approach only gives qualitatively correct results.

1.2.1 Second Order Phase Transitions

As seen in the phase diagram of second order phase transition (Fig 1.a), there is a symmetry in the theory upon changing variable from $-m \to m$. Thus, the free energy will only have even powers of m in its expansion.

$$\begin{aligned} F(T;m) &= F_0(T) + a(T)m^2 + b(T)m^4 + \dots && (17) \\ \frac{\partial F}{\partial m} &= 0 && (18) \\ \implies 2a(T)m + 4b(T)m^3 + \dots &= 0 && (19) \end{aligned}$$

Truncating at quartic order, the equation of state for m is approximately

$$2a(T)m + 4b(T)m^3 = 0 \tag{20}$$

Assuming $b(T) > 0 \forall T$, there are sketches of F corresponding to $a > 0$ and $a < 0$. They take

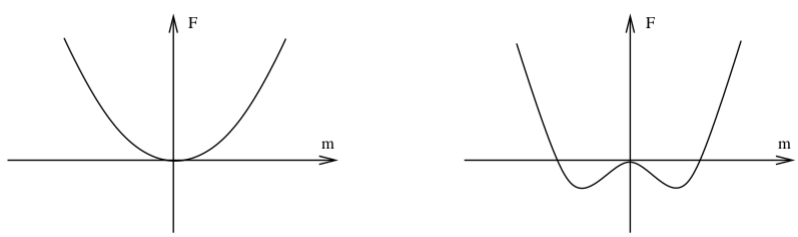

Figure 2: Free energy F for (a)LHS: $a(T) > 0$ (b)RHS: $a(T) < 0$

the same form as the free energy plots of the Ising model for $T > T_c$ and $T < T_c$ respectively. Comparing (20) with (7), we even get the correct critical exponent:

$$m_{Landau} \sim (T - T_c)^{\frac{1}{2}} \tag{21}$$

1.2.2 First Order Phase Transitions

The symmetry between $-m$ and m is lost here (as can be seen in Fig1.b). There will be odd powers of m in the expansion of F too. For $B \neq 0$, F_{Ising} too, takes a similar form.

$$\begin{aligned} F(T;m) &= F_0(T) + \alpha(T)m + a(T)m^2 + \gamma(T)m^3 + b(T)m^4 + \dots && (22) \\ F_{Ising}(T;m) &= -Nk_BTlog2 + \frac{JNq}{2}m^2 - \frac{N}{2k_BT}(B + Jqm)^3 + \dots && (23) \end{aligned}$$

Sketches of F_{Ising} for $B < 0, B = 0$ and $B > 0$ for small T are crudely similar to the sketches for a generic second order transition given by (22) for various conditions on α and γ.
We can clearly see the interpretations of the three different solutions for m:

STABLE $\to$ global minima of F
UNSTABLE $\to$ local maxima of F
METASTABLE $\to$ local, not global minima of F

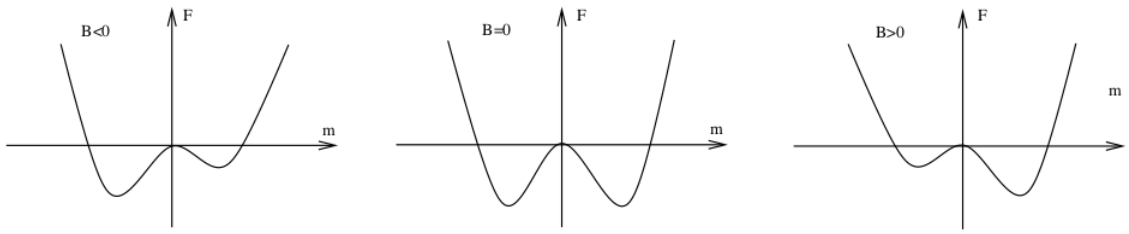

Figure 3: Free energy F of Ising Model for (a)LHS: $B < 0$ (b)Middle: $B = 0$ and (c)RHS: $B > 0$

At large temperatures, the kink disappears to give single minimum again. While the minimum is shifted from $m = 0$ for finite T, at extremely high temperatures, $m \to 0$ again.

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Apart from phase diagrams (Sec 1.1) and free energy distributions (Sec 1.2), another important characteristic of phase transitions is the notion of "critical slowing down".

1.3 Critical Slowing Down

Close to the critical points of a phase transition, if the system is perturbed away from equilibrium, it takes a very long time to decay back into the equibilrium state. This can be analysed for second order phase transitions more easily, in the Landau approach:

The Landau potential can be written as:

$$V(m) \approx \alpha(T - T_c)m^2 + \beta m^4 \tag{24}$$

At temperatures $T >> T_c$, the free energy distribution over m approaches a parabolic form, like the potential of a simple harmonic oscillator. However, we know that unlike a harmonic oscillator, the order parameter m never oscillates before reaching its steady state. We can therefore consider the dynamics of m reaching its equilibrium state to be like that of a harmonic oscillater *in the overdamped limit.*

In the overdamped limit, $\dot{x} = -\frac{dV}{dx} \implies$

$$\begin{aligned} V(m) &= \alpha(T - T_c)m^2 \\ \dot{m} &= -2\alpha(T - T_c)m \\ \implies m(t) &= m(0)e^{-\gamma(T)t} \quad \gamma(T) \sim T \end{aligned} \tag{25}$$

Thus, the relaxation back to equilibrium is exponentially fast, characterised by a temperature dependant time scale $\tau = 1/\gamma(T)$.

However, at $T \to T_c$, the quadratic term disappears:

$$\begin{aligned} V(m) &= \beta m^4 \\ \dot{m} &= 4\beta m^3 \\ \implies m(t) &\sim \frac{1}{\sqrt{\beta t}} \end{aligned} \tag{26}$$

Thus, the relaxation dynamics near T_c is now a power law decay which is considerably slower than (25).

We have discussed it for a second order phase transition because it is analogous to the overdamped simple harmonic oscillator - a familiar premise. However, we can generalise: phase transitions of the nth order will have a critically slow down, as given by a power law of $t^{(1/n)}$. Their dynamics are corresponding to other anharmonic oscillators in the overdamped limit (which enforces $\dot{x} = -\frac{dV}{dx}$).

2 Dynamical Systems and Bifurcation Point Behaviour

2.1 Introduction and Normal Forms

Any system (that evolves with time and) whose time evolution can be described by an analytical function for every point in phase space of a system, i.e. for $X \epsilon \mathbb{R}^N$

$$\dot{X} = F(X, \dot{X}, \ddot{X}, \ldots)$$

falls within the purview of dynamical systems theory.

A bifurcation refers to all the qualitative changes in the asymptotic dynamics observed when a parameter μ of the dynamics is varied. The value $\mu = \mu_0$ where the change occurs is called the *bifurcation point.*

In non-chaotic[5] systems, in the limit $t \to \infty$, every trajectory in the phase space is either at a fixed point or is in a periodic orbit (called a limit cycle); plotting these fixed points, x^* or the like as a function of the parameter μ gives us the *bifurcation diagram.*

There are many different types of bifurcations and each of them have a *normal form.* They are the form of the dynamics of every system that undergoes that particular bifurcation, near the bifurcation point, i.e. Regardless of the overall dynamics of a system, if it undergoes a particular kind of bifurcation

$\implies$ near its bifurcation point, the dynamics will reduce to corresponding the normal form

$\implies$ the normal form completely captures the behaviour of a system close to its bifurcation points.

Alternative approach to solving for fixed points

We can visualise dynamics of the first order system $\dot{x} = f(x)$ based on the idea of potential energy: by imagining the particle sliding down the walls of a potential well $V(x)$ in the overdamped limit (let us suppose there is some viscous coating on the surface of the well), where

$$f(x) = -\frac{dV}{dx}$$

[5]crudely, intuitively, this means that in the asymptotic limit, no trajectory will be wandering around in a bounded region without settling down to a fixed point or closed orbit. Chaotic systems are those where this *does* happen; the trajectories are attracted to a complex geometric object called a strange attractor. Their motion is aperiodic and enormously sensitive to changes in the initial conditions. They are studied differently

Now, by chain rule (and substituting $\frac{dx}{dt} = -\frac{dV}{dx}$)

$$\frac{dV}{dt} = \frac{dV}{dx}\frac{dx}{dt} \tag{27}$$

$$\frac{dV}{dt} = \frac{dV}{dx}\left(-\frac{dV}{dx}\right) \tag{28}$$

$$\frac{dV}{dt} = -\left(\frac{dV}{dx}\right)^2 \leq 0 \tag{29}$$

Thus the potential decreases, the particle always moves towards lower potential and will only reach equilibrium at a minima of V (where $\frac{dV}{dx} = 0$). Therefore, for a given first order system, if we can find a V such that:

$$V = \int_0^x -f(x')dx' \tag{30}$$

We can find equilibrium values for x. This is essentially a formulation of an analogue of Landau theory. The signs are already promising for a rich and deep connection between these two phenomena. Hereon in the text, this potential will be refered to as the bifurcation potential, U_{bif}.

We will now turn our attention to two fundamental types of bifurcations.

2.1.1 Subcritical Pitchfork Bifurcation

NORMAL FORM: $\dot{X} = -RX - X^3$ (or) $\dot{X} = RX + X^3$
BIFURCATION POTENTIAL: $U_{bif} \sim aX^2 + bX^4$

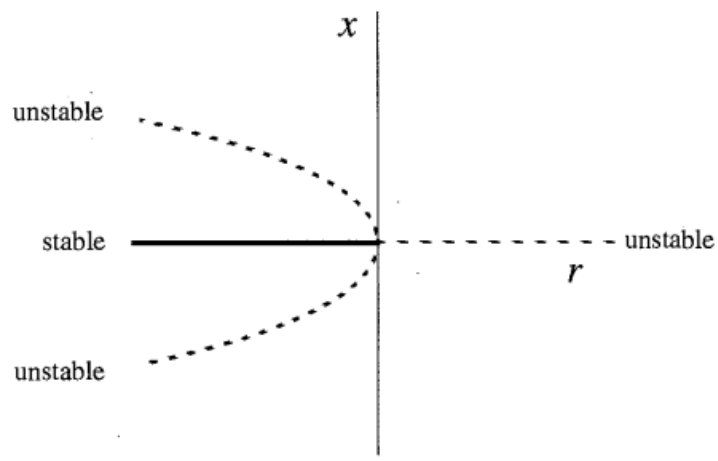

Figure 4: Bifurcation Diagram (Fixed Point Value x^* v/s parameter r). This is for the system $\dot{x} = rx + rx^3$ though we are interested in $\dot{x} = -(rx + rx^3)$, in which, the stable and unstable points are flipped.

CHARACTERISTICS:

1. There is a symmetry in the system, with no change in dynamics if $X \to -X$
2. One value of X is always a fixed point $\forall \mathbb{R}$. Beyond the bifurcation point, its nature changes (in the normal form, the fixed point $X = 0$ changes from unstable to stable at bifurcation point (at $R = 0$). 2 Symmetric values ($\pm\sqrt{-R}$) are stable fixed points for $R \leq 0$ but they cease to exist beyond bifurcation point.

MODIFICATION: The modified normal form

$$\dot{X} = -RX - X^3 + H \tag{31}$$

corresponds to an imperfect subcritical pitchfork bifurcation. While this complicates the form of the bifurcation diagram, in the limit $H \to 0$, all it does is merely shift the diagram along the parameter-axis. We can right away see the similarities between

1. Normal form for the imperfect case (31) and the equation of state for the Ising model (7) as well as for second order phase transition in Landau theory (20)
2. The bifurcation diagram in the limit $H \to 0$ (Fig.4) and the phase diagram for $B = 0$ case of Ising model (Fig.1.a).
3. The U_{bif} and the free energy of a second order phase transition using Landau theory (17).

2.1.2 Saddle Node Bifurcation

NORMAL FORM: $\dot{X} = R - X^2$ (or) $\dot{X} = R + X^2$
BIFURCATION POTENTIAL: $U_{bif} \sim aX + bX^3$

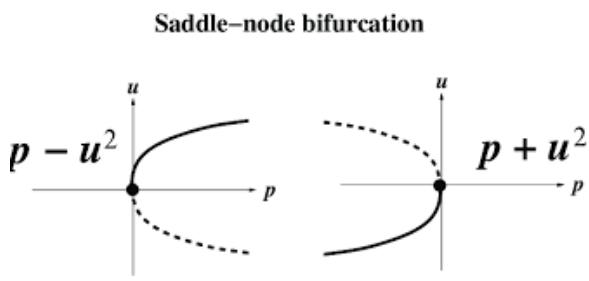

Figure 5: Bifurcation Diagram (Fixed Point Value u^* v/s parameter p) (a)LHS: $\dot{p} = p - u^2$ (b)RHS: $\dot{p} = p + u^2$

CHARACTERISTICS:

1. The number of fixed points is *not* conserved.
2. As R is varied, a pair of fixed points of opposite stability draw closer, eventually collide and annihilate. At the point of annihilation, a "ghost" is left behind: it is not a fixed point, but any trajectory passing through or close to it will take an inordinately long time to move past it.

MODIFICATION: If we consider the two different saddle node bifurcations corresponding to the two normal forms to coexist, with the bifurcation point of

$$\dot{X} = -(R + X^2) \text{ being } r_l$$
$$\dot{X} = R - X^2 \text{ being } r_u \text{ and } r_l < r_u$$

We will get the same bifurcation diagram as is sketched. This is actually analogous to the phase diagram for $B \neq 0$ case of the Ising model (Fig.1.b), in the limit $B \to 0$, for either sign of B. For a given value of temperature T, if we consider the bifurcation plot to be the phase diagram in

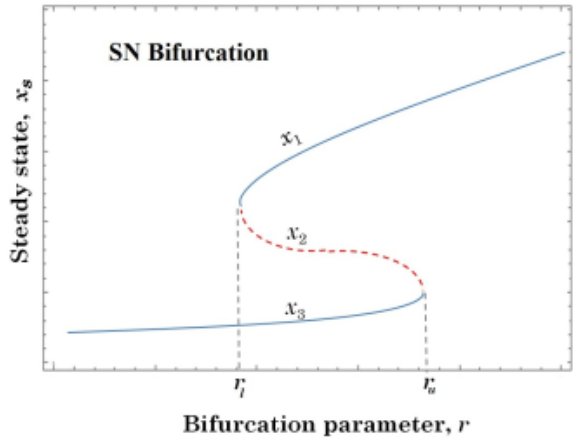

Figure 6:

greater resolution[6], we can now actually see the three solutions on each branch - the upper most corresponding to the stable solution (which persists $\forall T < T_c$), the middle being unstable and the lowermost being metastable respectively. The metastable state dies out as $T \ll T_c$, as expected.

Therefore, we can, yet again see the similarities and the consequences of the analogy between the modified saddle node bifurcation and a first order phase transition:

1. The bifurcation diagram (Fig.6) and the phase diagram (Fig.1.b) match (indeed, the former helps explain the latter better)

2. The U_{bif} and the free energy of a first order phase transition using Landau theory (22) are also comparable. (Both have a leading linear term. U_{bif} is missing the even terms, true, this is not the exact bifurcation potential for our modified saddle node bifurcation.)

2.2 Characterising Bifurcation Point Behaviour

Like in our treatment of phase transitions, critical behaviour of dynamical systems can be described not only by normal forms (equations of state) and bifurcation diagrams (phase diagrams) but also by an analogue to the **critical slowing down** observed in phase transitions. In the context of dynamical systems, the quantity is called the **divergence of characteristic time**.

2.2.1 Divergence of Characteristic Time

Consider a evolution $\dot{x} = V(\mu, x)$, with bifurcation point μ_0, with fixed point $x_0 = 0$ stable for $\mu < \mu_0$. Taylor expansion -

$$\dot{x} = V(x_0, \mu_0) + \left.\frac{\partial V}{\partial x}\right|_{x_0} (x - x_0) + \left.\frac{\partial^2 V}{\partial x^2}\right|_{x_0} (x - x_0)^2 + \dots \tag{32}$$

[6]In the coming section, we will aslo see why the experimentally obtained plots for Ising model are lacking in this level of resolution

For $\mu \ll \mu_0$, $\frac{\partial V}{\partial x}(x_0) < 0$, as x_0 is stable. Then:

$$\dot{x} \approx \left.\frac{\partial V}{\partial x}\right|_{x_0} (x - x_0) \tag{33}$$

$$\implies \dot{x} \sim -\frac{1}{\tau(\mu)}(x), \quad \tau(\mu) = \left|\frac{\partial V}{\partial x}(x_0)\right| \tag{34}$$

$$\implies x \sim e^{-\frac{t}{\tau}} \tag{35}$$

Thus, we see an exponential decay into the steady state for states far away from bifurcation point.

For $\mu = \mu_0$, as bifurcation occurs,

$$\frac{\partial V}{\partial x}(x_0, \mu_0) = 0 \tag{36}$$

$$\implies \tau(\mu_0) = \infty \tag{37}$$

$$\dot{x} \approx \left.\frac{\partial^2 V}{\partial x^2}\right|_{x_0} (x - x_0)^2 \tag{38}$$

$$\implies x(t) \sim \frac{1}{t} \tag{39}$$

Thus, $\tau(\mu_0)$, the characteristic time diverges at $\mu = \mu_0$. In general, for such an evolution, if $\left.\frac{\partial^k V}{\partial x^k}\right|_{x_0} = 0, \quad k = 1, 2, \ldots, n$ then,

$$x(t) \sim t^{\frac{-1}{n}} \tag{40}$$

We obtained a similar power law decay dynamics for phase transition of nth order.

NOTE When we were deriving the expression for U_{bif}, we assumed the system was similar to an overdamped (not necessarily harmonic) oscillator which was the same system we used to derive our critical slowing down power law for phase transitions.

3 Summing up the Analogies

As we can see, there are multiple threads of similarity, collated below.

Parameter	2nd order PT	SP Bifurcation
Normal form	$h - \theta m - \frac{1}{3}m^3 = 0$	$H - RX - X^3$
Landau Potential	$F(T; m) = F_0(T) + a(T)m^2 + b(T)m^4$	$U_{bif} \sim aX^2 + bX^4$
Parameter	1st order PT	SN Bifurcation
Landau Potential	$F(T; m) = F_0(T) + \alpha(T)m + a(T)m^2 + \gamma(T)m^3 + b(T)m^4$	$U_{bif} \sim aX + bX^3$
Parameter	n^{th} order PT	$**$
Critical Slowing Down	$m(t) \sim t^{(1/n)}$	$x(t) \sim t^{(1/n)}$

Where, $** = \left.\frac{\partial^k V}{\partial x^k}\right|_{x_0} = 0, \quad k \leq n$

There is correspondence in less quantitative terms as well:

1. Fairly obvious is the equivalence of m, order parameter and x, the dynamical variable. Thus, the steady state solutions to m also correspond to the fixed point values, x^*.

2. **Asymptotic Regime ($t \to \infty$) and Thermodynamic Limit ($N \to \infty$):** Both are idealisations that are never strictly reached. Moreover, these limits do not commute with singular parameters[7], e.g. In Ising model, $B_{ext} \to 0$ does not commute with $N \to \infty$, we must take B tending to 0 only after thermodynamic quantities have been calculated using the thermodynamic limit, else the obtained behaviours would differ. Renormalisation methods can correctly treat such singular systems.

3. **Finite Size Errors** As the sample for a phase transition will always have finite N, while in theory, phase transitions are only seen in thermodynamic limit, this leads to some blurring in the phase diagrams. Correspondingly, due to critical slowing down, the points observed near the bifurcation point have still not atttained their equilibrium position but are moving towards it too slowly; as the measurements are made at a finite time, we measure the non-equilibrium positions of these particles; thus the bifurcation diagrams too become more smoothed and blurry.

We sum up our qualitative discussion thus:

bifurcation	$\longleftrightarrow$	phase transition
t	$\longleftrightarrow$	r
duration T	$\longleftrightarrow$	size L
asymptotic regime	$\longleftrightarrow$	thermodynamic limit
divergence of characteristic time	$\longleftrightarrow$	critical slowing down

4 Importance and Consequences

The universality implied by the similarities in structure notwithstanding, of late there has been much work in extrapolating phase transitions to the dynamics of biological systems, leading to new insight.

Dynamics of biological system $\to$ phase transitions and analogous thermodynamic quantities

One such example is how Erez et al.(2019)[4] extended the MF Ising universality class of equilibrium thermodynamic models to include biochemical models with positive feedback describing nonequilibrium phenomena:

1. Using thermodynamic quantities (temperature T, magnetic field B, order parameter m and heat capacity) helped differentiate the responses of immune cells to different drugs in a simple way (just 3 parameters could characterise the responses completely). The standard way to characterise immune responses would involve trying to identify biological marker proteins and then running multiple assays measuring their levels - error prone, painstaking and with high dimensionality of data. The thermodynamic variables have unknowingly extracted the "principle components", have somewhat performed a PCA (Principal Component Analysis).

2. They also showed that the heat capacity allows inference of the absolute molecule number of the markers from fluorescence intensity.

[7] If ϵ is a singular parameter, it means the behaviour of the system is qualitatively different for $\epsilon = 0$ and otherwise.

5 Conclusion

This analogy between phase transitions and bifurcations is so profound that it has been called an identity - "a phase transition is nothing other than a bifurcation in the underlying microscopic dynamics"[3] of a system. Bifurcations are esssentially the underlying dynamics of phase transitions.

Moreoever, the study of critical phenomena in living systems is a newly emerging field, expected to witness a surge in activity in the coming years[5]. Thus, this not only has a rich theoretical basis but also very real applications and consequences.

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